CIRQUILL · HACKATHON 2026

PartLens

Evidence before confidence: AI-assisted electronics design where independent checks, not the model, decide.

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THE PROBLEM

A cheaper part can still break the board.

Electrical fit

Voltage, current, thermal limits and package all have to hold in this exact circuit.

Moving supply

Stock, minimum order and assembly fees change, and often go undocumented.

Confident AI

Chat assistants sound sure even when the evidence is missing.

PARTLENS

Two workflows, one honest answer.

New design from an idea

Requirements interview, reference architecture, generated parts list, supplier evidence and checks.

Existing BOM review

Match parts, check replacements, compare costs and recall earlier decisions.

Verified **Rejected** **Unverified** Missing evidence never becomes a yes.

HOW IT WORKS

The model investigates. Checks decide.

Local AI
qwen3.5:9b picks the next question or tool

Controller
Enforces steps, budgets and verbatim quotes

Evidence
JLCPCB API, Zetaris SQL, Meterless memory

Checks
Deterministic Python, exact Decimal maths

The report is rendered from tool results, not free text. The model can’t mark a part verified, invent a requirement or override a failed check.

LIVE DEMO · BREADBOARD POWER SUPPLY

The first regulator failed. PartLens caught it and switched.

XC6206 LDO · rejected

Current: 0.5005 A needed, 0.25 A limit
Heat: 0.976 W against a 0.25 W limit

TPS63031 buck-boost · clears both

The failed trial stays in the history with its recorded check results.

14 / 16
roles matched to official JLCPCB parts

2
shown as unmatched, never invented

Incomplete
cost: supplier terms undocumented

SPONSOR · ZETARIS

Supplier data, queried with audited SQL.

88

real read-only SELECTs in the official-source trial

- Lightning SQL through the Zetaris REST API, read-only
- Official JLCPCB observations queried as a relation, then tied to BOM lines and requirements
- Every query journalled: SQL, request ID, timing, rows, result digest
- No silent fallback: if Zetaris is down, the app says so

SPONSOR · METERLESS

Memory that has to earn its answer again.

3
recalled, still hold

2
recalled, still unverified

- Every decision stored in Meterless H-MEM with its run, checks and provenance
- Real reference engine in a separate Node process, bundled in the app
- Recall brings past decisions back as context, not as verdicts
- The controller forces fresh checks on current data before reuse

BUILT WITH

Private AI on an NVIDIA GPU. No cloud, no API credits.

NVIDIA GeForce RTX 5070 Ti

Ollama loads qwen3.5:9b (6.1 GB) fully into its 16 GB of GPU memory. Every agent turn runs locally on the GPU.

Development tools

Codex, Claude Code and Antigravity.
Cursor was not used. None of these tools run inside the product.

Paid cloud inference is disabled in the shipped app, and there is no silent fallback.

TRUST RULES, ENFORCED IN CODE

Only your own words confirm a requirement.

AI defaults stay provisional until you confirm them in your own words.

Exact Decimal maths for money and engineering values.

Your keys stay yours: encrypted per Windows user, never in the app.

The model never marks a part verified or computes a cost.

Nothing is deleted: failed trials and old runs stay on record.

392 automated tests guard these rules.

HONEST LIMITS · NEXT

Preliminary designs, clearly labelled.

- No validated schematic, ERC, layout or manufacturing files yet
- Next: schematic export, documented supplier terms, broader evaluation
- Declared pre-existing: Ollama and Qwen, Meterless H-MEM engine, Easyduino example BOM, open-source libraries

Try it

GitHub: github.com/zgeez/PartLens

Download: PartLens.exe from github.com/zgeez/PartLens/releases/latest

Video: PartLens-demo.mp4 from github.com/zgeez/PartLens/releases/latest

Start guide: docs/START-GUIDE.md